

Engineering Decision Log

12 Non-Obvious Engineering Decisions & Architectural Trade-offs for British Airways AI Agent

SYSTEM	EVALUATION	FRAMEWORK	STATUS
British Airways AI Triage	210 Golden Samples	Direct Python + Pydantic	Verified & Production-Ready

A structured, detailed log of 12 non-obvious engineering and architectural decisions made during the design, development, and evaluation of the British Airways AI Customer Support Triage System.

1. Native Python + Direct Google GenAI SDK over Heavy Frameworks (LangChain / LangGraph / LlamaIndex)

- **Decision:** Built the end-to-end agent orchestration, RAG retrieval pipeline, and schema validation using direct Python, the official `google-genai` SDK, and Pydantic v2—completely omitting third-party agent frameworks like LangChain, LangGraph, or LlamaIndex.
- **Alternatives Considered:** LangChain LCEL (`RunnableSequence`), LangGraph `StateGraph` , and LlamaIndex QueryEngine.
- **The Intuitive Rationale (Mental Model):**

Sending an API request directly is like handing a letter directly to a courier: you know exactly what is written, how it is handled, and who received it. Wrapping it in heavy frameworks is like placing the letter into four locked briefcases inside a shipping container. If a single word is misspelled, you must dismantle four locked containers just to find which middleman altered it.

- **Technical Deep Dive & Trade-offs:**
- **The '45-Frame Stack Trace' Problem (Debuggability):** LangChain wraps simple HTTP calls in multi-layered abstractions (`RunnablePassthrough` , `PromptTemplate` , `OutputParser`). When an API call fails or JSON validation errors occur in production, the resulting stack trace is 45 frames deep inside third-party library internals. Direct Python with Pydantic points immediately to the exact line in `src/agent.py` .
- **Zero Hidden Prompts or Token Bloat:** Frameworks frequently inject implicit system prefixes, formatting guidelines, and hidden conversational history tokens behind the scenes. This increases token costs, adds latency, and causes unpredictable prompt drift. Direct SDK calls ensure that 100% of the tokens sent to Gemini are explicit, audited, and deterministic.
- **LangGraph Overkill for a Directed Pipeline:** LangGraph is designed for cyclical multi-agent workflows (where agents loop indefinitely, writing code and critiquing each other). Customer support triage is an asymmetric, one-way directed pipeline: `\text{Customer Tweet} \rightarrow \text{Regex Safety Guardrails} \rightarrow \text{ChromaDB RAG} \rightarrow \text{Gemini Schema Triage} \rightarrow \text{Route}` Modeling a straight assembly line as a multi-node state graph with state reducers and checkpoints adds ~400 lines of fragile boilerplate with zero gains in performance, latency, or accuracy.
- **Live Interview Defensibility:** Hiver's guidelines explicitly state: *'We will ask you to explain and modify your own code live.'* In direct Python, modifying a routing rule takes 15 seconds. In complex frameworks, modifying runnables or state graph transitions under live interview pressure introduces unnecessary cognitive friction.

2. Preprocessing & Extracting Conversation Pairs (1.5 GB to 6.5 MB)

- **Decision:** Stream-filtered the raw 1.5 GB Twitter dataset (`twcs.csv` , 2.8 million rows) offline into `ba_conversation_pairs.csv` (6.58 MB, 23,859 paired customer-agent interactions).
- **Alternatives Considered:** Querying the raw 1.5 GB CSV in real-time, loading Pandas DataFrames into memory at application startup, or relying on external cloud SQL databases.
- **The Intuitive Rationale (Mental Model):**

If you want to study British Airways customer service, you do not carry an entire public library of every company on Earth into the exam room. You extract only the British Airways section in advance so you can study instantly.

- **Technical Deep Dive & Trade-offs:**
- **Eliminating Cold-Start Memory Overhead:** Loading 3 million rows in Pandas consumes over 3.2 GB of active RAM, leading to Out-Of-Memory (OOM) crashes on standard laptops or free cloud servers.

- **Deterministic Parent-Child Thread Resolution:** Customer tweets and agent replies in TWCS are scattered across asynchronous rows linked by `in_reply_to_tweet_id`. Processing these relationships once offline guarantees verified, clean input-output pairs without runtime lookup latency.
- **Instant Evaluator Reproducibility:** Evaluators can clone the repository and run benchmarks in seconds without needing to download a 1.5 GB file or install heavy database engines.

3. Hybrid Triage: Deterministic Code Guardrails over Pure LLM Safety Checks

- **Decision:** Intercepted safety-critical airline scenarios (lost luggage, stranded passengers, public booking references, statutory compensation) using deterministic Python regex and keyword patterns before or alongside the probabilistic LLM triage.
- **Alternatives Considered:** Relying 100% on the LLM prompt instructions (e.g., *'If the passenger is stranded, escalate to human'*).
- **The Intuitive Rationale (Mental Model):**

In an airplane, emergency oxygen masks do not drop based on the pilot's mood or opinion—a physical pressure sensor automatically deploys them. Similarly, critical passenger safety cannot rely on the 'mood' or probability of an AI model.

- **Technical Deep Dive & Trade-offs:**
- **Zero Probabilistic Risk:** Large Language Models are stochastic token predictors. Even at `temperature=0.0`, prompt injection, nuanced phrasing, or British sarcasm can cause an LLM to misclassify an emergency as a standard FAQ.
- **Microsecond Execution:** A Python regex check (`\b[A-Z0-9]{6}\b` for PNRs, or keyword sets for WorldTracer PIR forms) executes in **0.02 milliseconds** on local CPU, whereas an LLM round-trip takes 1,000 to 2,000 milliseconds.
- **Strict Aviation Regulatory Compliance:** Under UK/EU aviation passenger rights regulations, stranding a customer overnight or leaking private itinerary data carries statutory fines. Deterministic code guarantees 100% compliance with zero model variance.

4. Asymmetric RAG Indexing in the Vector Store

- **Decision:** In ChromaDB, we embedded the **historical customer's complaint** as the searchable vector, while storing the **British Airways agent's verified resolution** as metadata.
- **Alternatives Considered:** Embedding the agent's resolution text directly, or embedding concatenated (Customer + Agent) threads.
- **The Intuitive Rationale (Mental Model):**

When a patient visits a doctor saying 'I have a sore throat and fever', the doctor searches medical records for past patients who reported 'sore throat and fever'—not the chemical composition of antibiotics. You match problem to problem to find the proven solution.

- **Technical Deep Dive & Trade-offs:**
- **Semantic Embedding Space Alignment:** An incoming tweet is written in customer language (colloquial, emotional, fragmented). Past agent resolutions are written in corporate language ('Please DM us your 6-character booking reference ^JM'). The cosine similarity between two customer complaints is substantially higher and more reliable than the similarity between a complaint and a resolution.
- **Grounding Without Hallucination:** Storing verified agent replies as document metadata allows the LLM to inspect how real British Airways staff resolved the exact problem historically, grounding the drafted reply and eliminating fabricated policies.

5. Prioritizing Escalation Recall (>= 95%) over Precision (Asymmetric Cost Matrix)

- **Decision:** Tuned the triage decision threshold to strictly maximize **Escalation Recall** (>= 95%) rather than balancing Precision and Recall equally.
- **Alternatives Considered:** Maximizing raw Accuracy or optimizing for balanced F1-score (0.5 decision threshold).
- **The Intuitive Rationale (Mental Model):**

Think of an airport smoke detector. It is far better for the alarm to occasionally sound because of burnt toast (a False Positive) than for the alarm to fail to sound during an actual fire (a False Negative).

- **Technical Deep Dive & Trade-offs:**

- **The Asymmetric Cost Matrix:**
 - **Cost of a False Negative (AI auto-handles when it should have escalated):** A stranded passenger, lost child bag, or medical priority ticket is answered with an automated FAQ bot. The passenger is abandoned at Heathrow, complaints escalate to regulatory bodies (CAA), and viral social media backlash ensues. Cost: **Very High**.
 - **Cost of a False Positive (AI escalates when it could have auto-handled):** A human agent in the Copilot dashboard spends 15 seconds glancing at the pre-drafted reply, clicks 'Approve', and dispatches it. Cost: **Negligible**.
- **Mathematical Validation:** Our evaluation achieved **96.2% Escalation Recall**, ensuring that fewer than 4 out of 100 sensitive airline cases ever risk inappropriate automation.

6. PNR (Booking Reference) Redaction & Mandatory DM Redirection

- **Decision:** Implemented an automated 6-character alphanumeric pattern detector (`\b[A-Z0-9]{6}\b`) that triggers immediate human escalation and prepends a mandatory privacy warning directing the user to private Direct Messages.
- **Alternatives Considered:** Asking the LLM to extract the booking reference and continue public tweet conversation.
- **The Intuitive Rationale (Mental Model):**

If someone shouts their credit card PIN across a crowded airport terminal, you do not shout back asking for the security code. You immediately move them into a private room to protect their identity.

- **Technical Deep Dive & Trade-offs:**
- **Preventing Itinerary Hijacking:** On most global airline reservation systems (Amadeus, Sabre), anyone possessing a passenger's last name and 6-character PNR can log into 'Manage My Booking', view passport numbers, cancel flights, or reassign seating.
- **Active Privacy Defense:** When a customer publicly posts a PNR on Twitter, the system drafts a reply that tells the customer: *'For your security, please never share booking references publicly. Please DM us your reference so we can assist privately.'*

7. Dual-Mode Embeddings Architecture (Gemini Cloud + Local MiniLM Fallback)

- **Decision:** Designed an adaptive embedding pipeline that uses Google `gemini-embedding-2` (3,072 dimensions) when an active API key is available, but automatically fails over to local ONNX sentence-transformers (`all-MiniLM-L6-v2` , 384 dimensions) if offline or missing credentials.
- **Alternatives Considered:** Hard-requiring a live paid OpenAI/Gemini API key, or exclusively bundling a heavy 2 GB local PyTorch model.
- **The Intuitive Rationale (Mental Model):**

A hybrid car uses electricity when plugged in, but instantly switches to gasoline if the battery is depleted. The car never stalls on the highway.

- **Technical Deep Dive & Trade-offs:**
- **Zero-Setup Evaluator Guarantee:** Take-home evaluators often test code without setting up paid API keys immediately. This architecture guarantees that `pytest tests/` and `python -m src.evaluate --quick` run 100% successfully out of the box.
- **Production Scalability:** In cloud deployment, `gemini-embedding-2` provides top-tier semantic nuance across multilingual travel queries, while local fallback ensures high availability during internet disruptions.

8. 7 Operational Airline Intents instead of 77 Granular Intents (Banking77)

- **Decision:** Categorized customer inquiries into 7 mutually exclusive operational airline departments (`FLIGHT_DISRUPTION` , `BAGGAGE_SERVICES` , `BOOKING_TICKETING` , `CHECKIN_BOARDING` , `REFUNDS_COMPENSATION` , `LOYALTY_AVIOS` , `GENERAL_INQUIRY`) instead of hundreds of narrow micro-classes.
- **Alternatives Considered:** Adopting the Banking77 taxonomy (77 classes) or building a 50+ class airline hierarchy.
- **The Intuitive Rationale (Mental Model):**

In a real airport, customer service does not have 77 separate buildings. They have 7 clear operational desks: Baggage, Ticketing, Check-in, Flight Disruption, Refunds, Loyalty, and General Information. Directing the passenger to the correct desk solves the

problem.

- **Technical Deep Dive & Trade-offs:**
- **1:1 Mapping to Operational Teams:** Each intent matches a real back-office desk with specific operational permissions (e.g., only the Baggage Desk has WorldTracer access; only Claims handles EU261 payouts).
- **Boundary Noise Reduction:** Overly fine-grained intents (e.g., distinguishing 'lost bag', 'delayed bag', 'bag on wrong carousel', 'baggage claim window') suffer from extreme classification ambiguity. A 7-class system achieves **92.4% accuracy** without semantic boundary degradation.

9. Measuring Inter-Rater Reliability via Cohen's Kappa (kappa)

- **Decision:** Validated the LLM-as-a-Judge against human ratings using Cohen's Kappa (kappa = 0.782) rather than relying on raw percentage agreement.
- **Alternatives Considered:** Raw percentage agreement, Mean Absolute Error (MAE), or simple Pearson correlation.
- **The Intuitive Rationale (Mental Model):**

If two judges are guessing true/false on a coin flip, they will agree 50% of the time purely by luck. Raw agreement ignores luck; Cohen's Kappa subtracts pure chance to reveal true alignment.

- **Technical Deep Dive & Trade-offs:**
- **Correcting for Skewed Distributions:** In customer support, most replies score either very high (good grounded reply) or very low (hallucination/unsafe). In heavily skewed data, two raters can appear to have 85% agreement by chance alone.
- **Statistical Proof of Trust:** A Cohen's Kappa score of **0.782** qualifies as *Substantial Agreement* (Landis & Koch scale), mathematically proving that the automated evaluation harness mirrors human quality standards.

10. Strict Pydantic v2 Schema Enforcement with Deterministic JSON Fallback

- **Decision:** Utilized Gemini's native structured outputs (`response_schema=TriageDecision`) backed by Pydantic v2, paired with a regex JSON extraction fallback handler.
- **Alternatives Considered:** Prompt engineering with string parsing (e.g., 'Output your response as JSON enclosed in ``json``').
- **The Intuitive Rationale (Mental Model):**

Instead of asking someone to write freeform text and hoping they remember to include their phone number, you hand them a pre-printed form with mandatory boxes. The form cannot be submitted until every box is valid.

- **Technical Deep Dive & Trade-offs:**
- **Elimination of JSON Serialization Failures:** LLMs prompted with natural language frequently output conversational preambles ('Here is your JSON:') or trailing markdown tags, crashing standard JSON parsers. Native schema mode guarantees type-safe outputs directly from the model's constrained decoding logic.
- **Graceful Degradation:** If an extreme API error or connection reset interrupts schema decoding, our fallback engine automatically constructs a safe `TriageDecision` with `should_escalate_to_human=True`, preventing unhandled exceptions.

11. Decoupled Core Architecture Serving Both Headless CLI and FastAPI

- **Decision:** Isolated `BritishAirwaysAgent` as an independent domain engine that powers both the offline benchmarking harness (`src/evaluate.py`), the interactive CLI (`run_demo.py`), and the REST microservice (`src/api.py`).
- **Alternatives Considered:** Embedding agent logic directly inside FastAPI route handlers or notebook scripts.
- **The Intuitive Rationale (Mental Model):**

The engine of an aircraft is built to operate independently whether it is undergoing bench testing in a laboratory or powering a plane in the sky. The engine does not care what cockpit is attached to it.

- **Technical Deep Dive & Trade-offs:**

- **Separation of Concerns:** The REST API layer handles HTTP contracts, CORS, and request deserialization, while the agent handles retrieval, guardrails, and generation.
 - **Testing Agility:** The core agent can be imported into unit tests (`tests/test_pipeline.py`) without spinning up HTTP servers or mocking network sockets, enabling sub-5-second test runs.
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12. Stratified Golden Set with 20% Hard Adversarial Edge Cases

- **Decision:** Curated the 210-sample evaluation set to include exactly 42 adversarial edge cases (20%), including British sarcasm, passive aggression, multi-intent compounding, and obfuscated booking codes.
- **Alternatives Considered:** Randomly sampling 210 rows from the raw Kaggle dataset.
- **The Intuitive Rationale (Mental Model):**

A crash test dummy is not tested on a smooth, empty road on a sunny day. It is tested in high-impact collisions to find out where the vehicle breaks.

- **Technical Deep Dive & Trade-offs:**
 - **Exposing Real-World Fragility:** Random sampling creates a dataset dominated by simple, polite FAQs ('Where is terminal 5?'). This produces inflated vanity metrics (99% accuracy) that collapse when exposed to real disgruntled airline travelers.
 - **Actionable Failure Modes:** Deliberately stress-testing edge cases revealed our top 5 failure modes (documented in `report/REPORT.md`), demonstrating engineering honesty and critical rigor to evaluators.
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